

Bayesian Model(l)ing for Environmental Health

Welcome and Introduction

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Welcome!

Logistics

- **Wi-Fi Network:** guest-net (no password). Open any webpage on your browser, a pop-up will appear to connect to guest Wi-Fi. Accept terms to gain access.
- **Restrooms.** Directly outside the classroom to the right.
- **Course materials.** All material (lectures/labs) is located on Posit Cloud. You were invited to the 'sharp_bayesian_environmental_health_2025' workspace via email a few days ago.
- **Name tags.** Please wear during workshop to make connecting with others easier! Please return them after the Workshop to help us be greener.
- Contact Admin Coordinator Tristan Swedin ts3716@cumc.columbia.edu for any assistance.

Overview of workshop

The **Bayesian Modeling for Environmental Health Workshop** is a two-day intensive course of seminars and hands-on sessions to provide an *approachable* and *practical* overview of **concepts, techniques, and data analysis methods** used in Bayesian modeling with applications in Environmental Health.

Overview of workshop

- By the end of the workshop, participants should be familiar with the following topics:
 - Principles of Bayesian statistics
 - Bayesian workflow
 - Hierarchical modeling
 - Non-linear regression
 - Spatial and spatio-temporal modeling
 - Software options
 - Advanced models



Bayesian Modelling Workshop Team



Robbie M. Parks
Course Co-Director
Columbia University



Garyfallos Konstantinoudis
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Grace Santos
Workshop Fellow
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Nader Elmehdawi
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Special thank you



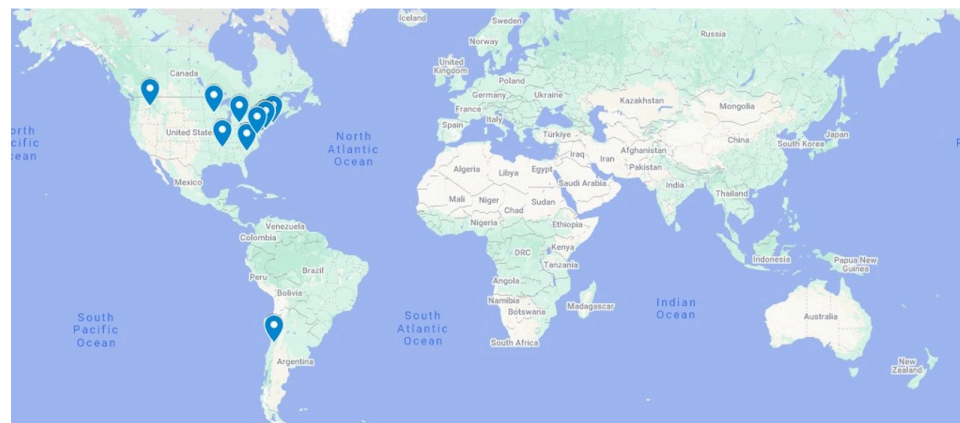
Theo Rashid
Instructor
Imperial College London



Elizaveta Semenova
Instructor
University of Oxford



13 US States + 1 country

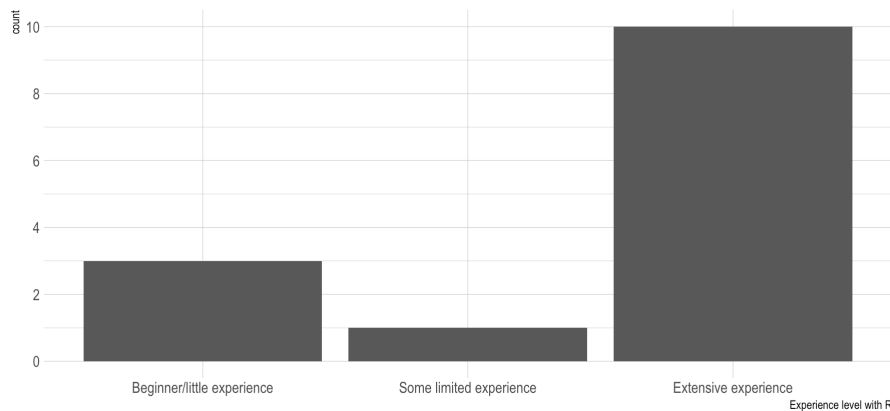


Day 1 (August 14th 2025)

Time	Activity
8:30 - 9:00	Check in and breakfast
9:00 - 9:15	Welcome and introduction
9:15 - 10:00	Principles of Bayesian statistics (Lecture)
10:00 - 10:15	Break
10:15 - 11:00	Principles of Bayesian statistics (Hands-on Lab)
11:00 - 11:15	Break
11:15 - 12:00	Bayesian workflow (Lecture)
12:00 - 1:00	Lunch
1:00 - 1:45	Hierarchical modeling (Lecture)
1:45 - 2:00	Break
2:00 - 2:45	Hierarchical modelling (Hands-on Lab)
2:45 - 3:00	Questions and wrap-up
3:00 - 3:45	Non-linear regression (Lecture)
3:45 - 4:00	Break
4:00 - 4:45	Non-linear regression (Hands-on Lab)
4:45 - 5:00	Questions and wrap-up



What is your experience level with R?



Day 2 (August 15th 2025)

Time	Activity
8:30 - 9:00	Check in and breakfast
9:00 - 10:00	Spatial and spatio-temporal modeling (Lecture)
10:00 - 10:15	Break
10:15 - 11:00	Spatial and spatio-temporal modelling (Hands-on Lab)
11:00 - 11:15	Break
11:15 - 12:00	Software options (Lecture)
12:00 - 1:00	Lunch
1:00 - 1:45	Software options (Hands-on Lab)
1:45 - 2:00	Break
2:00 - 2:45	Advanced models (Lecture)
2:45 - 3:00	Break
3:00 - 3:45	Advanced models (Hands-on Lab)
3:45 - 4:30	Interactive panel discussion & course wrap-up
4:30 - 5:00	Final farewell



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Questions?

